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Sheaf cohomology

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Introduction

Contents

Introduction i

1 Presheaves and sheaves 1
- 1.1 Presheaves . . . 1
- 1.2 Sheaves . . . 4

2 Ringed spaces and $\mathscr{O}_X$-modules. 9
- 2.1 Ringed spaces . . . 9
- 2.2 $\mathscr{O}_X$ modules . . . 9
- 2.3 $\mathscr{O}_X$-Mod is abelian. . . . 10
- 2.4 Sequences of $\mathscr{O}_X$-Mod . . . 12

A Category Theory 13

B Algebra 15

Chapter 1

Presheaves and sheaves

1.1 Presheaves

We begin by giving a general definition of a presheaf.

Definition 1.1.1 (Presheaf)**.** Let X be a topological space and let $\mathsf{Op}(X)$ be its Posetal category of open sets. Let C be a `def:small-category`. The Category of C-valued **presheaves** is the `ex:functor-category`

$$\mathsf{PSh}_X(\mathsf{C}) := [\mathsf{Op}(X)^{\mathrm{op}}, \mathsf{C}]$$

When the topological space is clear we will write $\mathsf{PSh}(\mathsf{C})$.

The elements of this category are Functors $\mathcal{F} : \mathsf{Op}(X)^{\mathrm{op}} \to \mathsf{C}$ and morphisms are Natural transformations $\phi : \mathcal{F} \to \mathcal{G}$. ⌟

Let's rewrite Definition 1.1.1 less abstractly: a C-valued Presheaf $\mathcal{F}$ is the collection of data

- For every open set $U \subseteq X$ an object $\mathcal{F}(U)$ of C called *sections of $\mathcal{F}$ over U*.
- For every inclussion of open sets $V \subseteq U$ a morphism $\mathrm{res}_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$ in C called *restriction* such that
  - For every open set U, $\mathrm{res}_{U,U} = \mathrm{id}_U$.
  - For every inclussion of open sets $W \subseteq V \subseteq U$, $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

A *morphism of Presheaves* $\phi : \mathcal{F} \to \mathcal{G}$ is a collection of morphisms $\{\phi(U) : \mathcal{F}(U) \to \mathcal{G}(U)\}_{U \in \mathsf{Op}(X)}$ in C making the following diagram commute:

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U) \\ \downarrow{\scriptstyle \mathrm{res}_{U,V}} & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ \mathcal{F}(V) & \xrightarrow{\phi(V)} & \mathcal{G}(V) \end{array}$$

We are going to study now Limits and Colimits in $\mathsf{PSh}(\mathsf{C})$.

Note that, since $\mathsf{PSh}(\mathsf{C})$ is a `def:functor-category` then if C has Limits or Colimits of a certain shape then $\mathsf{PSh}(\mathsf{C})$ also has them and they are computed pointwise.

This means that Limits and Colimits in $\mathsf{PSh}(\mathsf{C})$ can be computed *open set by open set*.

Proposition 1.1.1. *Let X be a topological space. Let $\mathcal{I} \to \mathsf{PSh}_X(\mathsf{C}), i \mapsto \mathcal{F}_i$ be a* `def:diagram`*. If C has Limits and/or Colimits with shape $\mathcal{I}$ then $\lim_i \mathcal{F}_i$ and/or $\operatorname{colim}_i \mathcal{F}_i$ exist in $\mathsf{PSh}_X(\mathsf{C})$. Moreover for every open set $U \subseteq X$*

$$\left(\lim_i \mathcal{F}_i\right)(U) = \lim_i \mathcal{F}_i(U) \qquad \left(\operatorname*{colim}_i \mathcal{F}_i\right)(U) = \operatorname*{colim}_i \mathcal{F}_i(U)$$

Write a section on functor categories (or quote it from somewhere) to justify this.

In particular if C is an Abelian category then Kernels and Cokernels exist and are computed *open set by open set*, making $\mathsf{PSh}(\mathsf{C})$ an Abelian category.

Remark 1.1.1. From this point onwards, we will suppose C has all Limits and Colimits unless stated otherwise. ▽

1.1.1 Presheaves and continuous maps

Definition 1.1.2 (Direct image Presheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathcal{F}$ be a C-Presheaf over X. The **direct image Presheaf** of $\mathcal{F}$ via f is the Presheaf $f_*\mathcal{F}$ defined as, for every open subset $V \subseteq Y$

$$f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(U))$$

Given $W \subseteq V \subseteq Y$ open subsets, the restriction map comes from the commutativity of the diagram

$$\begin{array}{ccc} f_*\mathcal{F}(V) & = & \mathcal{F}(f^{-1}V) \\ \vdots & & \downarrow \\ f_*\mathcal{F}(V) & = & \mathcal{F}(f^{-1}V) \end{array}$$

This construction defines a Functor $f_* : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{PSh}_Y(\mathsf{C})$. ⌟

Show that the category of presheaves over a one point space *is equal* to the evaluation category.

Example 1.1.1 (Global section)**.** Let X be a topological space. Consider the one-point topological space $\{*\}$. Let $\mathcal{F}$ be a C-valued presheaf over X. The **global section** of $\mathcal{F}$ over

This defines a Functor $\Gamma(X, -) : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{C}$ ◊

Example 1.1.2 (Skyscrapper Presheaf)**.** Let X be a topological space and $p \in X$. Suppose C has a `def:final-object` $* \in \mathsf{C}$. Let S be an object in C. The **skyscrapper Presheaf** supported at S $\iota_{p,*}\underline{S}_{\text{pre}}$ is the Direct image Presheaf of the Constant Presheaf $\underline{S}_{\text{pre}}$ via the inclussion $\iota : \{p\} \hookrightarrow X$:

$$(\iota_{p,*}\underline{S}_{\text{pre}})(U) = \begin{cases} S & p \in U \\ * & p \notin U \end{cases}$$

◊

Using this we can construct a *dual* Presheaf to Definition 1.1.2.

Definition 1.1.3 (Inverse image Presheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathcal{G}$ be a C-valued Presheaf over Y. The **inverse image Presheaf** of $\mathcal{G}$ via f is the Presheaf $f^{-1}\mathcal{G}$ defined as, for every open subset $U \subseteq X$

$$f^{-1}\mathcal{G}(U) = \operatorname*{colim}_{V \supseteq f(U)} \mathcal{G}(V)$$

where the Colimit is taken over all the open neighbourhoods V of $f(U)$.

This defines a Functor $f^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$. ⌟

Remark 1.1.2. Note that the set of open neighbourhoods of $f(U)$ is a partially ordered set via reverse inclussion. Therefore if C is a concrete category we can describe $f^{-1}\mathcal{G}(U)$ as $\{(s, V) \mid f(U) \subseteq V \in \mathsf{Op}(Y), s \in \mathcal{G}(V)\}$ modulo the relation $(s, V) \sim (s', V')$ if and only if there exists $W \subseteq V \cap V'$ open with $f(U) \subseteq W$ such that $s|_W = s'|_W$. ▽

Example 1.1.3 (Constant Presheaf)**.** Let X be a topological space. Let S be an object of C. The **constant Presheaf** associated to S $\underline{S}_{\text{pre}}$ is the Inverse image Presheaf of S via the map $\{*\} \to X$. ◊

Definition 1.1.4 (Stalk)**.** Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued Presheaf over X. The **stalk** of $\mathcal{F}$ at p $\mathcal{F}_p$ is the Inverse image Presheaf of $\mathcal{F}$ via $X \to \{p\}$:

$$\mathcal{F}_p = \operatorname*{colim}_{U \ni p} \mathcal{F}(U)$$

⌟

Proposition 1.1.2. *Let $f : X \to Y$ be a continuous map. We have an* `def:adjunction` *between the Direct image Presheaf functor and the Inverse image Presheaf $f^{-1} \dashv f_*$.*

Proof. We define the `def:categorical-unit` $\eta : 1 \implies f_* f^{-1}$ as the map with components given by the canonical insertion maps to the colimit

$$\mathcal{F}(W) \to \operatorname*{colim}_{U \supseteq (f \circ f^{-1})(W)} \mathcal{F}(U) = f^{-1}\mathcal{F}(f^{-1}W) = f_* f^{-1}\mathcal{F}(W)$$

Note that this maps exist because $(f \circ f^{-1})(W) \subseteq W$.

On the other hand, note that if $f(V) \subseteq U$ then $V \subseteq (f^{-1} \circ f)(V) \subseteq f^{-1}(U)$, i.e: we have a restricion map $\mathcal{G}(f^{-1}(U)) \to \mathcal{G}(V)$. Therefore by the universal property of the colimit we have a unique map

$$f^{-1} f_* \mathcal{G}(V) = \operatorname*{colim}_{U \supseteq f(V)} \mathcal{G}(f^{-1}(U)) \to \mathcal{G}(V)$$

Using this maps we obtain a `def:categorical-counit` $\varepsilon : f^{-1} f_* \implies 1$.

Since by construction the compositions $\eta \circ \varepsilon$ and $\varepsilon \circ \eta$ are the corresponding identity morphisms, we have proven the adjunction. □

Two important particular cases of the adjunction Proposition 1.1.2 are

Corollary 1.1.1. *Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued Presheaves over X and let A be an object in C. Then*

$$\mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\underline{A}_{\text{pre}}, \mathcal{F}) \simeq \mathrm{Hom}_{\mathsf{C}}(A, \Gamma(X, \mathcal{F}))$$

Corollary 1.1.2. *Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued Presheaves over X and let A be an object in C. Then*

$$\mathrm{Hom}_{\mathsf{C}}(\mathcal{F}_p, A) \simeq \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota_{p,*}\underline{A}_{\text{pre}})$$

1.2 Sheaves

We now turn our attention to a particular class of Presheaves, following [**stacks-project**].

Let's start by considering the case $\mathsf{C} = \mathsf{Set}$:

Definition 1.2.1 (Sheaf of sets)**.** Let X be a topological space. A Presheaf of sets $\mathcal{F}$ is a **sheaf** if for every open set U and open covering $\{U_i\}_{i\in\mathcal{I}}$ of U the diagram

$$\mathcal{F}(U) \to \prod_{i\in\mathcal{I}} \mathcal{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathcal{F}(U_i \cap U_j) \tag{1.2.1}$$

is an Equalizer.

We denote by $\mathsf{Sh}_X(\mathsf{Set})$ (or just Sh if the context is clear) the Full Subcategory of the category of presheaves of sets whose objects are sheaves of sets.

Less abstractly this means that $\mathcal{F}$ verifies:

SH.1) **Identity**: Given two sections $s, t \in \mathscr{F}(U)$ such that for every $i \in \mathcal{I}$ $\quad s|_{U_i} = t|_{U_i}$ we have that $s = t$.

SH.2) **Gluability**: Given a collection of sections $\{s_i \mid s_i \in \mathscr{F}(U_i)\}_{i\in\mathcal{I}}$ that agree on pairwise intersections ($\forall i, j \in \mathcal{I}\ \ s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$) there exists a section $s \in \mathscr{F}(U)$ such that $\forall i \in \mathcal{I}\ \ s|_{U_i} = s_i$.

⌟

Definition 1.2.2 (Category of Sheaves of sets)**.** The Category $\mathsf{Sh}_X(\mathsf{Set})$ of Sheaves of sets is the Full Subcategory of the category of Presheaves $\mathsf{PSh}_X(\mathsf{Set})$ whose objects are Sheaves of sets. ⌟

Let's try to make Definition 1.2.1 more general by considering another evaluation category C.

Firstly, note that if Products indexed by $\mathcal{I}$ and $\mathcal{I}\times\mathcal{I}$ for any indexing set $\mathcal{I}$ exist in C then the diagram (1.2.1) makes sense. Secondly, we have that (1.2.1) is an Equalizer if and only if for every object $C \in \mathsf{C}$ the diagram in Set

$$\mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U)) \to \prod_{i\in\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U_i)) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U_i \cap U_j))$$

is an Equalizer.

Therefore we get

Definition 1.2.3 (Sheaf)**.** Let X be a topological space. Let C be an arbitrary category. A C-valued Presheaf $\mathcal{F}$ is a **sheaf** if for all objects $C \in \mathsf{C}$ the Presheaf of sets $\mathcal{F}_C$ defined as $\mathcal{F}_C(U) = \mathrm{Hom}_{\mathsf{C}}(C, \mathcal{F}(U))$ is a Sheaf of sets.

We denote by $\mathsf{Sh}_X(\mathsf{C})$ the Full Subcategory of the category of C-valued presheaves of sets whose objects are C-valued sheaves. ⌟

Notation. From now on, to distinguish a Presheaf from a Sheaf we will denote the former by $\mathcal{F}, \mathcal{G}, \mathcal{H}, \ldots$ and the latter by $\mathscr{F}, \mathscr{G}, \mathscr{H}, \ldots$

By Definition 1.2.3 we have an inclussion functor $\iota : \mathsf{Sh}(\mathsf{C}) \hookrightarrow \mathsf{PSh}(\mathsf{C})$.

For *sufficiently nice* Categorys, this Functor admits a left-adjoint $\mathsf{PSh}(\mathsf{C}) \to \mathsf{Sh}(\mathsf{C})$. Let's study first the case $\mathsf{C} = \mathsf{Set}$.

How general do we want to go?

Theorem 1.2.1 (Sheafification). *There exists a* ***sheafification*** *Functor* $(-)^{\#} : \mathsf{PSh}(\mathsf{Set}) \to \mathsf{Sh}(\mathsf{Set})$ *that is left-adjoint to the inclussion Functor* $\iota : \mathsf{Sh}(\mathsf{Set}) \to \mathsf{PSh}(\mathsf{Set})$.

Proof. Let $\mathcal{F}$ be a Presheaf of sets over X. For every open subset $U \subseteq X$ we define

$$\mathcal{F}^{\#}(U) = \left\{ f_p \in \prod_{p \in U} \mathcal{F}_p \text{ verifying } (*) \right\}$$

where $(*)$ is the property *"for all $p \in U$ there exists an open neighbourhood $V \subseteq U$ of p and a section $s \in \mathcal{F}(V)$ such that, for every $q \in V$, $s_q = f_q$"*.

Taking $\prod_{p \in U} \mathcal{F}_p \to \prod_{q \in V} \mathcal{F}_q$ as the restriction maps we see that $\mathcal{F}^{\#}$ is a Presheaf. Let's see now that $\mathcal{F}^{\#}$ is a Sheaf. To do this, consider an open subset $U \subseteq X$ with open cover $\{U_i\}_{i \in \mathcal{I}}$.

SH.1 Let $s, t \in \mathcal{F}^{\#}(U)$ be two sections such that, for all $i \in \mathcal{I}$ $s|_{U_i} = t|_{U_i}$.

Then since for every $p \in U$ there exists $i \in \mathcal{I}$ such that $p \in U_i$ we have that $s_p = (s|_{U_i})_p = (t|_{u_i})_p = t_p$. Therefore $s = t$.

SH.2 Let $\{s_i \in \mathcal{F}^{\#}(U_i)\}_{i \in \mathcal{I}}$ be a collection of sections such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in \mathcal{I}$.

Consider $s \in \prod_{p \in U} \mathcal{F}_p$ defined as $s_p = (s_i)_p$ if $p \in U_i$. Note that this is well-defined by property $(*)$. To see that s verifies $(*)$ note that for every $p \in U_i$ there exists an open neighbourhood $V \subseteq U$ of p and $t \in \mathcal{F}(V)$ such that, for all $q \in V$, $t_q = (s_i)|_q = s_q$.

Let $V \subseteq U \subseteq X$ be two open sets. Consider the commutative diagram

$$\begin{array}{ccccc} \mathcal{F}(U) & \longrightarrow & \mathcal{F}^{\#}(U) & \longrightarrow & \prod_{p \in U} \mathcal{F}_p \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{F}(V) & \longrightarrow & \mathcal{F}^{\#}(V) & \longrightarrow & \prod_{q \in V} \mathcal{F}_q \end{array}$$

where the vertical arrows are the corresponding restriction maps. By chasing this diagram we get a map $\varphi : \mathcal{F} \to \mathcal{F}^{\#}$, $\varphi_U(s) = (s_p)_{p \in U}$.

It only remains to show that any morphism $\psi : \mathcal{F} \to \mathcal{G}$, where $\mathcal{G}$ is a Sheaf of sets, factors uniquely as $\mathcal{F} \xrightarrow{\varphi} \mathcal{F}^{\#} \to \mathcal{G}$, i.e: we have to construct a morphism $\theta : \mathcal{F}^{\#} \to \mathcal{G}$ such that $\psi = \theta \circ \varphi$.

Let $U \subseteq X$ be an open subset and $s \in \mathcal{F}^{\#}(U)$. For each $p \in U$ consider an open neighbourhood $V^{(p)} \subseteq U$ of p and $t^{(p)} \in \mathcal{F}(V^{(p)})$ such that $t^{(p)}_q = s_q$. Consider now $\psi_{V^{(p)}}(t^{(p)}) \in \mathcal{G}(V^{(p)})$. Note that $\{\psi_{V^{(p)}}(t^{(p)})\}_{p \in U}$ is a compatible family of section in $\mathcal{G}$, therefore by the Sheaf axioms there exists a unique section $t \in \mathcal{G}(U)$ such that $t|_{V^{(p)}} = \psi_{V^{(p)}}(s^{(p)})$. Therefore we can define $\theta_U(s) = t$.

By construction $\psi = \theta \circ \varphi$ and θ commutes with the restriction maps, making it a morphism of sheaves.

To check uniqueness, let $\theta' : \mathcal{F}^{\#} \to \mathcal{G}$ be another morphism such that $\psi = \theta' \circ \varphi$. Consider $s \in \mathcal{F}^{\#}(U)$. For each $p \in U$ choose $V^{(p)} \subseteq U$ and $s^{(p)} \in \mathcal{F}(V^{(p)})$ such that $s|_{V^{(p)}} = \varphi_{V^{(p)}}(s^{(p)})$. Then

$$\theta'(s)|_{V^{(p)}} = \theta'(s|_{V^{(p)}}) = \theta'(\varphi_{V^{(p)}}(s^{(p)})) = \psi_{V^{(p)}}(s^{(p)})$$

On the other hand $\theta(s)|_{V^{(p)}} = \psi_{V^{(p)}}(s^{(p)})$ by definition. Therefore since $\mathscr{G}$ is a Sheaf by SH.1 we have $\theta'(f) = \theta(f)$, i.e: $\theta' = \theta$. □

We can extend the construction outlined in Theorem 1.2.1 to Algebraic Categorys.

Lemma 1.2.1. *Let (C, F) be an Algebraic Category. Let $\mathcal{F}$ be a C-valued Presheaf. Then there exists a C-valued Sheaf $\mathcal{F}^{\#}$ and a morphism of presheaves $\mathcal{F} \to \mathcal{F}^{\#}$ satisfying*

1. *The map $\mathcal{F} \to \mathcal{F}^{\#}$ identifies the underlying Sheaf of sets of $\mathcal{F}^{\#}$ with the sheafification of the underlying Presheaf of sets of $\mathcal{F}$.*
2. *For any morphism of presheaves $\mathcal{F} \to \mathscr{G}$ where $\mathscr{G}$ is a C-valued Sheaf there exists a unique factorization $\mathcal{F} \to \mathcal{F}^{\#} \to \mathscr{G}$.*

Remark 1.2.1. We will assume that C is an Algebraic Category in what follows. ▽

Therefore we can assign a Sheaf to any Presheaf using this construction.

Definition 1.2.4 (Sheafification)**.** Let $\mathcal{F}$ be a C-valued Presheaf. We call the Sheaf $\mathcal{F}^{\#}$ obtained by Theorem 1.2.1 the **sheafification** of $\mathcal{F}$. ⌟

Using this adjunction we can *adapt* Definition 1.1.2 and Definition 1.1.3 to the case of Sheaves

Definition 1.2.5 (Direct image Sheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathscr{F}$ be a C-valued Sheaf over X. The **direct image Sheaf** of $\mathscr{F}$ via f is the Sheaf $f_*\mathscr{F}$ defined as, for every open subset $V \subseteq Y$

$$f_*\mathscr{F}(V) = \mathscr{F}(f^{-1}(U))$$

This construction defines a Functor $f_* : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{Sh}_Y(\mathsf{C})$. ⌟

This means that Example 1.1.1 and Example 1.1.2 translate directly to Sheaves.

Definition 1.2.6 (Global section Functor)**.** We denote by $\Gamma(X, -) : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$ the **global section functor** described in Example 1.1.1, but now with initial category $\mathsf{Sh}_X(\mathsf{C})$. ⌟

Definition 1.2.7 (Skyscrapper Sheaf)**.** Let X be a topological space and $p \in X$. Let S be an object of C. The **skyscrapper Sheaf** supported at p $\iota_{*,p}\underline{S}$ is the Direct image Sheaf of the Constant Sheaf $\underline{S}$ via the map $\{p\} \to X$. ⌟

Rewrite this definitions

Explain why we need to sheafificate

Definition 1.2.8 (Inverse image Sheaf)**.** Let $f : X \to Y$ be a continuous map. Let $\mathscr{G}$ be a C-valued Sheaf over Y. The **inverse image Sheaf** of $\mathscr{G}$ via f is the Sheaf $f^{-1}\mathscr{G}$ defined as, for every open subset $U \subseteq X$

$$f^{-1}\mathscr{G} = (f^{-1}\mathscr{G})^{\#}$$

where the RHS of the equation is the Sheafification of the Inverse image Presheaf of $\mathscr{G}$ via f.

This defines a Functor $f^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$. ⌟

Definition 1.2.9 (Constant Sheaf)**.** Let S be an object of C. The **constant Sheaf** $\underline{S}$ associated to S is the Sheafification of the Constant Presheaf $\underline{S}_{\text{pre}}$, i.e: $\underline{S} = (\underline{S}_{\text{pre}})^{\#}$. ⌟

Definition 1.2.10 (Stalk)**.** Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued Sheaf over X. The **stalk** of $\mathscr{F}$ at p $\mathscr{F}_p$ is the Inverse image Sheaf of $\mathscr{F}$ via $X \to \{p\}$:

$$\mathscr{F}_p = \operatorname*{colim}_{U \ni p} \mathscr{F}(U)$$

⌟

Using the adjointness we have the analogous of Proposition 1.1.2

Proposition 1.2.1. *Let $f : X \to Y$ be a continuous map. We have an* `def:adjunction` *between the Direct image Sheaf functor and the Inverse image Sheaf $f^{-1} \dashv f_*$.*

Proof.

$$\begin{aligned}\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(f^{-1}\mathscr{G}, \mathscr{F}) &= \operatorname{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(f^{-1}\mathscr{G}, \mathscr{F}) \\ &= \operatorname{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathscr{G}, f_*\mathscr{G}) \\ &= \operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{G}, f_*\mathscr{G})\end{aligned}$$

□

Therefore adjunctions Corollary 1.1.1 and Corollary 1.2.2 also happen when considering Sheaves.

Corollary 1.2.1. *Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued Sheaves over X and let A be an object in C. Then*

$$\operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\underline{A}, \mathscr{F}) \simeq \operatorname{Hom}_{\mathsf{C}}(A, \Gamma(X, \mathscr{F}))$$

Corollary 1.2.2. *Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued Sheaves over X and let A be an object in C. Then*

$$\operatorname{Hom}_{\mathsf{C}}(\mathscr{F}_p, A) \simeq \operatorname{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{F}, \iota_{p,*}\underline{A})$$

For cohomology we will be interested in the case $\mathsf{C} = \mathsf{Ab}$.

Lemma 1.2.2. *Let $a : \mathsf{A} \to \mathsf{B}$ and $b : \mathsf{B} \to \mathsf{A}$ be two Functors. Suppose that*

1. *A and B are additive categories, a, b are additive functors and $b \dashv a$.*
2. *B is an Abelian category and b is left-exact.*
3. *$ba \cong \mathrm{id}_{\mathsf{A}}$.*

Then A is abelian.

Proof. Since B is an Abelian category all Limits and Colimits exist en B.

Since b is left-adjoint it is right-exact and therefore, exact.

Let $\varphi : B_1 \to B_2$ be a morphism in B. If $K = \ker(\varphi)$ then K is the Equalizer of the diagram

$$K \longrightarrow B_1 \underset{0}{\overset{\varphi}{\rightrightarrows}} B_2$$

therefore applying b we get

$$bK \longrightarrow bB_1 \underset{0}{\overset{b\varphi}{\rightrightarrows}} bB_2$$

i.e: bK is the Kernel of $b\varphi$.

Dually we get that $b\operatorname{coker}(B_1 \to B_2)$ is the Cokernel of $b\varphi$.

Therefore we have that every morphism $b\varphi$ in A admits a Kernel and a Cokernel. Moreover since $ba \cong \mathrm{id}_{\mathsf{A}}$ every morphism in A is of the form $b\varphi$, therefore all Kernels and Cokernels exist in A. Indeed for every $\psi : A_1 \to A_2$ we have that

$$\ker \psi = b \ker(a\psi) \qquad \operatorname{coker} \psi = b \operatorname{coker}(a\psi) \tag{1.2.2}$$

It only remains to show that $\operatorname{coim} \psi = \operatorname{im} \psi$. Since A has all Limits and Colimits A has all finite Limits and Colimits, making a left-exact. This means that a transforms Kernels into Kernels. Therefore

$$\begin{aligned}
\operatorname{coim} \psi &= \operatorname{coker}(\ker \psi \to A_1) && \text{(definition)} \\
&= b \operatorname{coker}(a(\ker \psi \to A_1)) && \text{(1.2.2)} \\
&= b \operatorname{coker}(\ker(a\psi) \to aA_1) && (a \text{ preserves kernels}) \\
&= b \operatorname{coim}(a\psi) && \text{(definition)} \\
&= b \operatorname{im}(a\psi) && (\mathsf{B} \text{ is abelian}) \\
&= b \ker(aA_2 \to \operatorname{coker}(a\psi)) && \text{(definition)} \\
&= \ker(baA_2 \to b \operatorname{coker}(a\psi)) && (b \text{ preserves kernels}) \\
&= \ker(A_2 \to b \operatorname{coker}(a\psi)) && ba \cong \mathrm{id}_{\mathsf{A}} \\
&= \ker(A_2 \to \operatorname{coker} \psi) && \text{(1.2.2)} \\
&= \operatorname{im} \psi && \text{(definition)}
\end{aligned}$$

□

Proposition 1.2.2. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism in $\mathsf{Sh}(\mathsf{Ab})$.*

1. *The Category $\mathsf{Sh}_X(\mathsf{Ab})$ is an Abelian category.*
2. *The Kernel of φ is the same as the Kernel of φ in $\mathsf{PSh}(\mathsf{Ab})$.*
3. *φ is injective if and only if φ is injective as a map or presheaves if and only if φ is injective as a map of sheaves.*
4. *The Cokernel of φ is the Sheafification of the Cokernel of φ in $\mathsf{PSh}(\mathsf{Ab})$.*
5. *φ is surjective if and only if φ is surjective as a map of sheaves.*
6. *A complex $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact at $\mathscr{G}$ if and only if for all open sets U and sections $s \in \mathscr{G}(U)$ mapping to 0 in $\mathscr{H}(U)$ there exists an open covering $\{U_i\}_{i\in\mathcal{I}}$ such that each $s|_{U_i}$ is the image of $\mathscr{F}(U_i) \to \mathscr{G}(U_i)$.*

Proof. Note that 2. $\implies$ 3., 4. $\implies$ 5 and 2. $\wedge$ 5. $\implies$ 6.

Therefore we only need to prove 1., 2. and 3., but this comes directly from Lemma 1.2.2 with $\mathbf{A} = \mathsf{Sh}_X(\mathsf{Ab})$, $\mathsf{B} = \mathsf{PSh}_X(\mathsf{Ab})$, $a : \mathsf{Sh}_X(\mathsf{Ab}) \hookrightarrow \mathsf{PSh}_X(\mathsf{Ab})$ the inclussion Functor and $b : \mathsf{PSh}_X(\mathsf{Ab}) \to \mathsf{Sh}_X(\mathsf{Ab})$ the Sheafification Functor. □

Chapter 2

Ringed spaces and $\mathscr{O}_X$-modules.

2.1 Ringed spaces

Definition 2.1.1 (Ringed space)**.** A **ringed space** is a tuple $(X, \mathscr{O}_X)$ where X is a topological space and $\mathscr{O}_X$ is a Sheaf of rings over X.

We call $\mathscr{O}_X$ the *structure sheaf* of the ringed space $(X, \mathscr{O}_X)$. Sections of $\mathscr{O}_X$ over an open subset $U \subseteq X$ are called *functions on* U and functions on X are called (global) functions. ⌟

Example 2.1.1. Let and X be a topological space. Then the $\mathbb{R}$-Algebra of continuous functions $\mathcal{C}^0_X(U) = \{f : U \to \mathbb{R} \mid f \text{ continuous.}\}$ with the obvious restriction maps forms a Sheaf of rings over X. Therefore $(X, \mathcal{C}^0_X)$ is a Ringed space. ◊

Example 2.1.2. If for every point $p \in X$ the **??**s $\mathscr{O}_{X,p}$ of the structure sheaf are Local rings then we call $(X, \mathscr{O}_X)$ a **locally ringed space**. ◊

Definition 2.1.2 (Morphism of Ringed spaces)**.** Let $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ be two Ringed spaces. A **morphism of ringed spaces** $(X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a pair $(f, f^\sharp)$ where $f : X \to Y$ is a continuous function (i.e: a morphism in Top) and $f^\sharp : \mathscr{O}_Y \to f_*\mathscr{O}_X$ is a def:sheaf-morphism of rings (i.e: a morphism in $\mathrm{Sh}_{\mathrm{Ring}}(X)$), where $f_*\mathscr{O}_X$ is the def:pushfo eaf. ⌟

Maybe it is better to define $f^\sharp$ as an f-map. **[stacks-project]**

Example 2.1.3. Let $f : X \to Y$ be a continuous function. Consider the Ringed spaces of continuous functions $(X, \mathcal{C}^0_X)$, $(Y, \mathcal{C}^0_Y)$.

Then the Morphism of Ringed spaces induced by f can be defined for every $V \subseteq Y$ open as $f^\sharp(V) : \mathcal{C}^0_Y(V) \to \mathcal{C}^0_X(f^{-1}(V))$, $f^\sharp(V)(h) = h \circ f|_{f^{-1}V}$. ◊

2.2 $\mathscr{O}_X$ modules

Definition 2.2.1 ($\mathscr{O}_X$-module)**.** Let $(X, \mathscr{O}_X)$ be a Ringed space. A **Sheaf of $\mathscr{O}_X$ modules** (or simply an $\mathscr{O}_X$-module) is a Sheaf $\mathscr{F}$ such that for every open set $U \subseteq X$ $\mathscr{F}(U)$ is an $\mathscr{O}_X(U)$-Module compatible with the restriction maps, i.e: for every $f \in \mathscr{O}_X(U)$ and $s \in \mathscr{F}(U)$: $(f \cdot s)|_V = f|_V \cdot s|_V$. ⌟

Example 2.2.1. ... ◊

Definition 2.2.2 (Morphism of $\mathscr{O}_X$-modules)**.** Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules. A morphism $\phi : \mathscr{F} \to \mathscr{G}$ is a def:sheaf-morphism that commutes with the multiplication of the structure sheaf $\mathscr{O}_X$. ⌟

Example 2.2.2. ... ◊

Therefore for every Ringed space $(X, \mathscr{O}_X)$ the collection of $\mathscr{O}_X$-modules forms a Category $\mathscr{O}_X$-Mod.

2.2.1 $\mathscr{O}_X$-Mod and sheaves of abelian groups.

Just as abelian groups are $\mathbb{Z}$-modules, Sheaves of abelian groups can be seen as $\underline{\mathbb{Z}}_X$-modules, where $\underline{\mathbb{Z}}_X$ is the `ex:constant-sheaf` with coefficients in the integers.

To show this, for every Sheaf of abelian groups $\mathscr{F}$ over X and every open set $U \subseteq X$ we have to define an action $\cdot : \underline{\mathbb{Z}}_X(U) \times \mathscr{F}(U) \to \mathscr{F}(U)$ compatible with the restriction maps.

Let $f \in \underline{\mathbb{Z}}_X(U)$ and $s \in \mathscr{F}(U)$. Since f is locally constant there exists an open cover $\{U_i\}_{i\in\mathcal{I}}$ of U such that for every $i \in \mathcal{I} \quad f|_{U_i} = n_i \in \mathbb{Z}$.

For every U_i consider $t_i = n_i \cdot s|_{U_i}$, where $n_i \cdot s|_{U_i}$ is shorthand for $s|_{U_i}$ added to itself n_i times. Note that this is well-defined because $\mathscr{F}(U_i)$ is an abelian group.

Let $U_i, U_j \in \{U_i\}_{i\in\mathcal{I}}$ such that $U_i \cap U_j = \varnothing$. For every $x \in U_i \cap U_j$ there exists an open neighbourhood $V \subseteq U_i \cap U_j$ of x were f is constant.

Therefore $n_i = n_j$ and $t_i|_{U_i\cap U_j} = t_j|_{U_i\cap U_j}$ and, applying axiom SH.2, we can define $f \cdot s = t$.

Checking the axioms in small enough open sets such that f is constant verifies that this action makes $\mathscr{F}$ an $\underline{\mathbb{Z}}_X$-module.

Do we have to actually write the check?

2.3 $\mathscr{O}_X$-Mod is abelian.

Definition 2.3.1 (Preadditive category)**.** A (small) category C is **preadditive** if every hom-set $\mathrm{Hom}(\triangle, \square)$ admits the structure of an abelian group and composition of morphisms is bilinear with respect to the group operation. ⌟

Definition 2.3.2 (Additive category)**.** A category C is **additive** if it is a Preadditive category, has a zero object and has all binary biproducts. ⌟

Definition 2.3.3 (Abelian category)**.** A Additive category C is **abelian** if

- It has all kernels and cokernels.
- Every monomorphism is the kernel of some morphism and every epimorphism is the cokernel of some morphism.

⌟

Move this definitions to the Homological Algebra section

Let's check first that $\mathscr{O}_X$-Mod is a Preadditive category.

Lemma 2.3.1. *The category $\mathscr{O}_X$-Mod is a Preadditive category.*

Proof. Let $\phi : \mathscr{F} \to \mathscr{G}$ and $\psi : \mathscr{F} \to \mathscr{G}$ be two Morphisms of Ringed spaces.

For each open subset $U \subseteq X$ define $(\phi + \psi)(U) = \phi(U) + \psi(U)$. □

Now Additive category

Lemma 2.3.2. *The category $\mathscr{O}_X$-Mod admits a zero object.*

Proof. Consider the $\mathscr{O}_X$ module that sends every open set to $\{0\}$. □

Lemma 2.3.3. *The category $\mathscr{O}_X$-Mod has all binary biproducts. We define their direct sum a $\mathscr{F} \oplus \mathscr{G} = \mathscr{F} \times \mathscr{G}$.*

Proof. Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules. □

Theorem 2.3.1. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ a Morphism of $\mathscr{O}_X$-modules. Then the kernel of φ is the subsheaf of $\mathscr{F}$ with sections*

$$(\ker \varphi)(U) = \{s \in \mathscr{F}(U) \mid \varphi(s) = 0\}$$

Furthermore for every $x \in X$ $\quad (\ker \varphi)_x = \ker \varphi_x$.

Proof. If $V \subseteq U$

$$\begin{array}{ccccccc} 0 & \longrightarrow & (\ker\varphi)(U) & \longrightarrow & \mathscr{F}(U) & \xrightarrow{\varphi(U)} & \mathscr{G}(U) \\ & & \vdots & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ 0 & \longrightarrow & (\ker\varphi)(V) & \longrightarrow & \mathscr{F}(V) & \xrightarrow{\varphi(V)} & \mathscr{G}(V) \end{array}$$

□

Theorem 2.3.2. *Let $\varphi : \mathscr{F} \to \mathscr{G}$ a Morphism of $\mathscr{O}_X$-modules. Then the cokernel of φ is the Sheafification of the quotient Presheaf*

$$(\mathrm{coker}_{\mathrm{pre}}\, \varphi)(U) = {}^{\mathscr{G}(U)}\big/_{\varphi(\mathscr{F}(U))}$$

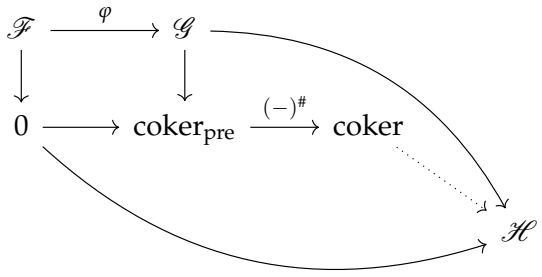
Prove that this is the cokernel in PSh.

Furthermore for every $x \in X$ $\quad (\mathrm{coker}\, \varphi)_x = \mathrm{coker}\, \varphi_x$.

Proof. Given a commutative diagram

$$\begin{array}{ccc} \mathscr{F} & \xrightarrow{\varphi} & \mathscr{G} \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathscr{H} \end{array}$$

We construct



□

Therefore we have proven that $\mathscr{O}_X$-Mod is an Abelian category and in particular by $\mathscr{O}_X$-Mod and sheaves of abelian groups $\mathsf{Sh}_{\mathsf{Ab}}(X)$ is an Abelian category.

2.4 Sequences of $\mathscr{O}_X$-Mod

Theorem 2.4.1. *The sequence of $\mathscr{O}_X$-modules $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact at $\mathscr{G}$ if and only if for every $x \in X$ the sequence $\mathscr{F}_x \to \mathscr{G}_x \to \mathscr{H}_x$ is exact at $\mathscr{G}_x$.*

Probably not a *theorem* but a direct consequence of the commutatitivity of stalks and (co)kernels

Maybe talk about resolutions / types of resolutions here?

Appendix A

Category Theory

Definition A.0.1 (Category). ⌟

Example A.0.1 (Posetal category). ◊

Example A.0.2 (Posetal category of open sets). ◊

Definition A.0.2 (Functor). ⌟

Definition A.0.3 (Natural transformation). ⌟

Definition A.0.4 (Limit). ⌟

Definition A.0.5 (Colimit). ⌟

Definition A.0.6 (Equalizer). ⌟

Definition A.0.7 (Adjoint). ⌟

Definition A.0.8 (Product). ⌟

Definition A.0.9 (Coproduct). ⌟

Definition A.0.10 (Full Functor). ⌟

Definition A.0.11 (Faithful Functor). ⌟

Definition A.0.12 (Fully faithful Functor). ⌟

Definition A.0.13 (Reflects). ⌟

Definition A.0.14 (Isomorphim). ⌟

Definition A.0.15 (Subcategory). ⌟

Definition A.0.16 (Full Subcategory). ⌟

Definition A.0.17 (Directed Colimit). ⌟

Proposition A.0.1 (Functors preserve Isomorphims).

Definition A.0.18 (Universal property). ⌟

Lemma A.0.1 (Yonneda lemma).

Definition A.0.19 (Algebraic Category). ⌟

Move

Definition A.0.20 (Abelian category)**.** ⌟

Definition A.0.21 (Kernel)**.** ⌟

Definition A.0.22 (Cokernel)**.** ⌟

Appendix B

Algebra

Definition B.0.1 (Algebra)**.** ⌟

Definition B.0.2 (Local ring)**.** ⌟

Definition B.0.3 (Module)**.** ⌟

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